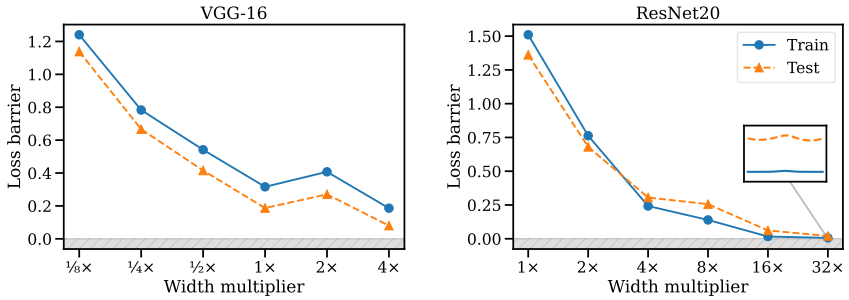

Figure 4: **Wider models exhibit better linear mode connectivity.** Training convolutional and ResNet architectures on CIFAR-10, we ablate their width and visualize loss barriers after weight matching. Notably, we achieve zero-barrier linear mode connectivity between ResNet models, the first such demonstration.

respectively. We then evaluate slices through the loss landscape, $\mathcal{L}((1-\lambda)\Theta_A + \lambda\pi(\Theta_B))$ for $\lambda \in [0, 1]$, where π is selected according to the methods presented in Section 3.[3] Ideally, we seek a completely flat or even convex one-dimensional slice. As discussed in Section 2, the ability to exhibit this behavior for arbitrary Θ_A, Θ_B empirically suggests that the loss landscape contains only a single basin modulo permutation symmetries.

We remark that a failure to find a π such that linear mode connectivity holds cannot rule out the existence of a satisfactory permutation. Given the astronomical number of permutation symmetries, Conjecture 1 is essentially impossible to disprove for any realistically wide model architecture.

5.1 Loss Landscapes Before and After Matching

We present results for models trained on MNIST (LeCun et al., 1998), CIFAR-10 (Krizhevsky, 2009), and ImageNet (Deng et al., 2009) in Figure 2. Naïve interpolation ($\pi(\Theta_B) = \Theta_B$) substantially degrades performance when interpolating. On the other hand, the methods introduced in Section 3 can achieve much better barriers. We achieve zero-barrier linear mode connectivity on MNIST with all three methods, although activation matching performs just slightly less favorably than weight matching and straight-through estimator (STE) matching. We especially note that the test loss landscape becomes convex after applying our weight matching and STE permutations! In other words, our interpolation actually yields a merged model that outperforms both models A and B. We elaborate on this phenomenon in Section 5.4 and Appendix A.10.

On ImageNet we fall short of zero-barrier connections, although we do see a 67% decrease in barrier relative to naïve interpolation. As we demonstrate in Section 5.3, we can achieve zero-barrier LMC on CIFAR-10 with large ResNet models. Therefore, we hypothesize that the presence of LMC depends on the model having sufficient capacity (esp. width) to capture the complexity of the input data distribution, and that ImageNet results could be improved by expanding model width.

STE matching, the most expensive method, produces the best solutions. Somewhat surprising, however, is that the gap between STE and the other two methods is relatively small. In particular, it is remarkable how well Algorithm 1 performs without access to the input data at all. We found that weight matching offered a compelling balance between computational cost and performance: It runs in mere seconds (on current hardware) and produces high-quality solutions.

5.2 Onset of Mode Connectivity

Given the results of Section 5.1, it may be tempting to conclude that the entirety of weight space contains only a single basin modulo permutation symmetries. However, we found that linear mode connectivity is an emergent property of training, and we were unable to uncover it early in training. We explore the emergence of LMC in Figure 3. Concurrent to our work, Benzing et al. (2022) showed that LMC at initialization is possible using a permutation found at the end of training.

[3] We also experimented with spherical linear interpolation ("slerp") and found it to perform slightly better than linear interpolation in some cases; however, the difference was not sufficiently significant to warrant diverging from the pre-existing literature.

Note that the final inlaid interpolation plot in Figure 3(right) demonstrates an important shortcoming of the loss barrier metric, i.e., the interpolation includes points with lower loss than either of the two models. However, the loss barrier is still positive due to non-negativity, as mentioned in Section 2.

5.3 Effect of Model Width

Conventional wisdom maintains that wider architectures are easier to optimize (Jacot et al., 2018; Lee et al., 2019). We now investigate whether they are also easier to linearly mode connect. We train VGG-16 (Simonyan & Zisserman, 2015) and ResNet20 (He et al., 2016) architectures of varying widths on the CIFAR-10 dataset. Results are presented in Figure 4.[4]

A clear relationship emerges between model width and linear mode connectivity, as measured by the loss barrier between solutions. Although $1\times$-sized models did not seem to exhibit linear mode connectivity, we found that larger width models decreased loss barriers all the way to zero. In Figure 4(right), we show what is to our knowledge the premiere demonstration of zero-barrier linear mode connectivity between two large ResNet models trained on a non-trivial dataset.

We highlight that relatively thin models do not seem to obey linear mode connectivity yet still exhibit similarities in training dynamics. This suggests that either our permutation selection methods are failing to find satisfactory permutations on thinner models or that some form of invariance other than permutation symmetries must be at play in the thin model regime.

5.4 Model Patching, Split Data Training, and Improved Calibration

Inspired by work on finetuning (Wortsman et al., 2022a), model patching (Singh & Jaggi, 2020; Raffel, 2021), and federated learning (McMahan et al., 2017; Konečný et al., 2016a;b), we study whether it is possible to synergistically merge the weights of two models trained on disjoint datasets. Consider, for example, an organization with multiple (possibly biased) datasets separated for regulatory (e.g., GDPR) or privacy (e.g., on-device data) considerations. Models can be trained on each dataset individually, but training in aggregate is not feasible. Can we combine separately trained models so that the merged model performs well on the entirety of the data?

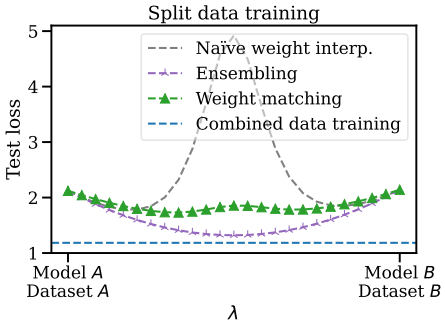

Figure 5: **Models trained on disjoint datasets can be merged constructively.** Algorithm 1 makes it possible for two ResNet models trained on disjoint, biased subsets of CIFAR-100 to be merged in weight space such that their combination outperforms both input models in terms of test loss on the combined dataset.

To address this question, we split the CIFAR-100 dataset (Krizhevsky, 2009) into two disjoint subsets: dataset A, containing 20% examples labelled 0-49 and 80% labelled 50-99, and dataset B, vice versa. ResNet20 models A and B were trained on their corresponding datasets. Privacy requirements mandate that we utilize a data-agnostic algorithm like Algorithm 1. Figure 5 shows the result of merging the two models with weight matching. For comparison, we benchmark naïve weight interpolation, ensembling of the model logits, and full-data training.

As expected, merging separately trained models did not match the performance of an omniscient model trained on the full dataset or an ensemble of the two models with twice the number of effective weights. On the other hand, we did manage to merge the two models in weight space, achieving an interpolated model that outperforms both input models in terms of test loss while using half the memory and compute required for ensembling. Furthermore, the merged model's probability estimates are better calibrated than either of the input models as demonstrated in Figure 11. Accuracy results are presented in Figure 10. Algorithm 1 also vastly outperformed naïve interpolation, the status quo for model combination in federated learning and distributed training.

[4]Unfortunately, $8\times$ width VGG-16 training was unattainable since it exhausted GPU memory on available hardware at the time of writing.

6 Related Work

(Linear) mode connectivity. Garipov et al. (2018); Draxler et al. (2018); Freeman & Bruna (2017) showed that different solutions in the neural network loss landscape could be connected by paths of near-constant loss, which Garipov et al. (2018) coined "mode connectivity." Tatro et al. (2020) explored non-linear mode connectivity modulo permutation symmetries. Frankle et al. (2020) demonstrated a connection between *linear* mode connectivity and the lottery ticket hypothesis. Juneja et al. (2022) demonstrated that LMC does not always hold, even when fine-tuning. Hecht-Nielsen (1990); Chen et al. (1993) noted the existence of permutation symmetries, and Brea et al. (2019) implicated them as a source of saddle points in the loss landscape. Recently, the prescient work of Entezari et al. (2021) conjectured that SGD solutions could be linear mode connected modulo permutation symmetries and offered experiments buttressing this claim. Unlike previous works on LMC we accomplish zero-barrier paths between two independently-trained models with an algorithm that runs on the order of seconds.

Loss landscapes and training dynamics. Li et al. (2016); Yosinski et al. (2014) investigated whether independently trained networks learn similar features, and to what extent they transfer. Jiang et al. (2021) argued that independently trained networks meaningfully differ in the features they learn in certain scenarios. Zhang et al. (2019) studied the relative importance of layers. Benton et al. (2021) argued that SGD solutions form a connected volume of low loss. Pittorino et al. (2022) proposed a toroidal topology of solutions and a set of algorithms for symmetry removal. On the theoretical front, Kawaguchi (2016) proved that deep linear networks contain no local minima. Boursier et al. (2022); Chizat & Bach (2018); Mei et al. (2018) characterized the training dynamics of one-hidden layer networks, proving that they converge to zero loss. Godfrey et al. (2022); Simsek et al. (2021) investigated the algebraic structure of symmetries in neural networks and how this structure manifests in loss landscape geometry.

Federated learning and model merging. McMahan et al. (2017); Konečný et al. (2016a;b) introduced the concept of "federated learning," i.e., learning split across across multiple devices and datasets. Wang et al. (2020) proposed an exciting federated learning method in which model averaging is done after permuting units. Unlike this work, they merged smaller "child" models into a larger "main" model, and did so with a layer-wise algorithm that does not support residual connections or normalization layers. Raffel (2021); Matena & Raffel (2021); Sung et al. (2021) conceptualized the study of "model patching," i.e., the idea that models should be easy to modify and submit changes to. Ilharco et al. (2022) investigated model patching for the fine-tuning of open-vocabulary vision models. Ashmore & Gashler (2015) first explored the use of matching algorithms for the alignment of network units. Singh & Jaggi (2020) proposed merging models by soft-aligning associations weights, inspired by optimal transport. Liu et al. (2022a); Uriot & Izzo (2020) further explored merging models taking possible permutations into account. Wortsman et al. (2022a) demonstrated state-of-the-art ImageNet performance by averaging the weights of many fine-tuned models.

7 Discussion and Future Work

We explore the role of permutation symmetries in the linear mode connectivity of SGD solutions. We present three algorithms to canonicalize independent neural network weights in order to make the loss landscape between them as flat as possible. In contrast to prior work, we linearly mode connect large ResNet models with no barrier in seconds to minutes. Despite presenting successes across multiple architectures and datasets, linear mode connectivity between thin models remains elusive. Therefore, we conjecture that permutation symmetries are a necessary piece, though not a complete picture, of the fundamental invariances at play in neural network training dynamics. In particular, we hypothesize that linear, possibly non-permutation, relationships connect the layer-wise activations between models trained by SGD. In the infinite width limit, there exist satisfactory linear relationships that are also permutations.

An expanded theory and empirical exploration of other invariances – such as cross-layer scaling or general linear relationships between activations – presents an intriguing avenue for future work. Ultimately, we anticipate that a lucid understanding of loss landscape geometry will not only advance the theory of deep learning but will also promote the development of better optimization, federated learning, and ensembling techniques.